
Training on Augmented Reality

TRAINING REPORT

Submitted in partial fulfillment of the requirements for the award of the degree

Of

-BACHELOR OF TECHNOLOGY-

In

-Electronics and Communication Engineering-

By

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-07101022019-

Guided by

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-Bolt IoT-



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NEW DELHI – 110006**

- JULY 2021 -

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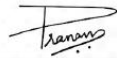


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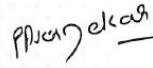
This is to certify that Anushka Singhania, a student of Indira Gandhi Delhi Technical University for Women has scored 25 marks out of 25 and successfully completed the 1 Month training on Augment Reality.

09/08/2021

Date



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Anushka Singhania

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ACKNOWLEDGEMENT

I am thankful to ‘BoltIot’ and my course instructor ‘Sachin Soni’ for carrying out this online course on Training on Augmented Reality.

I am also thankful to my college “IGDTUW” for encouraging us to learn this course.

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DECLARATION

I, **Anushka Singhania** solemnly declare that the project report, **TRAINING ON AUGMENTED REALITY** is based on my own work carried out during the course of our study under the supervision of **Sachin Soni, Bolt IoT**. I assert the statements made and conclusions drawn are an outcome of my research work. I further certify that:

- I. The work contained in the report is original and has been done by me under the supervision of my supervisor.
- II. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.
- III. We have followed the guidelines provided by the university in writing the report.
- IV. Whenever we have used materials (text, data, theoretical analysis/equations, codes/program, figures, tables, pictures, text etc.) from other sources, we have given due credit to them in the report and have also given their details in the references.

Anushka Singhania

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LIST OF ABBREVIATIONS

Abbreviation	Description
AR	Augmented Reality
VR	Virtual Reality
HMD	Head Mounted Display
GPS	Global Positioning System
FPS	Frame per second

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ABSTRACT/SUMMARY

Augmented reality (AR) is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. It is a growing trend among companies involved in mobile computing and business applications in particular. This Training on Augmented Reality focuses What is AR, how is it different from VR, its application in real world and making of projects from scratch using Spark AR studio by Facebook.

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CHAPTER 1: INTRODUCTION

Technology has advanced to the point where realism in virtual reality is very achievable. However, in our obsession to reproduce the world and human experience in virtual space, we overlook the most important aspects of what makes us who we are-our reality. Yet, it isn't enough just to trick the eye or fool the body and mind. One must capture the imagination in order to create truly compelling experiences.

On the spectrum between VR, which creates immersible, computer-generated environments, and the real world, augmented reality is closer to the real world. AR adds graphics, sounds, haptics and smell to the natural world as it exists. You can expect video games to drive the development of augmented reality, but this technology will have countless applications. Everyone from tourists to military troops will benefit from the ability to place computer-generated graphics in their field of vision.

AR will truly change the way we view the world. Picture yourself walking or driving down the street. With augmented-reality displays, which will eventually look much like a normal pair of glasses, informative graphics will appear in your field of view and audio will coincide with whatever you see. These enhancements will be refreshed continually to reflect the movements of your head. In this article, we will take a look at this future technology, its components and how it will be used.

With the introduction of AR as being coined the term in the early nineties, we were able to apply virtual objects within physical reality. Combining techniques Sutherland/Sproul's first optical see-through HMD from the early 1960's with complex, real-time computer-generated wiring diagrams and manuals. Both were registered with each other and manuals were embedded within the actual aircraft for intensely detailed procedures.

AR refers to computer displays that add virtual information to a user's sensory perceptions. Most AR research focuses on see-through devices, usually worn on the head that overlay graphics and

text on the user's view of his or her surroundings. In general it superimposes graphics over a real world environment in real time.

Getting the right information at the right time and the right place is the key in all these applications. Personal digital assistants such as the Palm and the Pocket PC can provide timely information using wireless networking and GPS receivers that constantly track the handheld devices. But what makes augmented reality different is how the information is presented: not on a separate display but integrated with the user's perceptions. This kind of interface minimizes the extra mental effort that a user has to expend when switching his or her attention back and forth between real-world tasks and a computer screen. In augmented reality, the user's view of the world and the computer interface literally become one.

AR is far more advanced than any technology you've seen in television broadcasts, although early versions of augmented reality are starting to appear in televised races and football games. These systems display graphics for only one point of view. Next-generation augmented reality systems will display graphics for each viewer's perspective.

CHAPTER 2: AUGMENTED REALITY

2.1 What is Augmented Reality?

AR is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. It is a growing trend among companies involved in mobile computing and business applications in particular.

Amid the rise of data collection and analysis, one of augmented reality's primary goals is to highlight specific features of the physical world, increase understanding of those features, and derive smart and accessible insight that can be applied to real-world applications. Such big data can help inform companies' decision-making and gain insight into consumer spending habits, among others.

Augmented reality continues to develop and become more pervasive among a wide range of applications. Since its conception, marketers and technology firms have had to battle the perception that augmented reality is little more than a marketing tool. However, there is evidence that consumers are beginning to derive tangible benefits from this functionality and expect it as part of their purchasing process.

2.2 AR v/s VR

AR	VR
• AR uses a real-world setting	• VR is completely virtual
• AR users can control their presence in the real world	• VR users are controlled by the system
• AR can be accessed with a smartphone	• VR requires a headset device
• AR enhances both the virtual and real world	• VR only enhances a fictional reality

Table 1: AR v/s VR

CHAPTER 3: SPARK AR

3.1 What is Spark AR?

Spark AR is a studio tool from Facebook that allows users to create their own AR effects for mobile. First launched in 2017, Facebook continues to add capabilities to the platform – most recently, adding in analytics for Instagram and Facebook campaigns. Compatible for Mac and Windows, the AR platform is comparable to tools such as Sketch or Photoshop – only this is for augmented reality.

3.2 What Spark AR can do?

Spark AR enables you to create your own AR effects for mobile using a suite of tools – from patching to animation. What’s more, you don’t have to be a technical genius to be able to use it. It allows you to:

- Create your own AR effects, with or without a technical background
- Import your own 3D files and sounds
- Build with or without code

With a host of features and learning guides, it’s a good tool for professional creators (such as studios like ours), as well as personal users. For example, Spark AR enables you to insert your own 3D objects into a project, change their properties and add interactivity, logic and animation. However, if you don’t have your own 3D models, Spark AR provides a vast range of ready-made 3D objects in the AR library.

3.3 Why Spark AR?

Platforms like Spark AR demonstrate how mixed-reality technology is becoming more mainstream – more accessible, useful and fun. Spark AR enables you to create anything from a face filter to interactive AR games, and social media gives it a wide-ranging audience. Meanwhile, the importance of marketing via social media goes without saying. Instagram has over 500 million daily users and is ever-growing in popularity, with advertisers reporting four times as much engagement on Instagram as Facebook.

Spark AR enables brands to be both more creative and more engaging, while taking advantage of detailed metrics, including total impressions, captures and shares – giving insights into how their audience responds to AR. It shows how the technology is evolving – both in terms of ease and capability, as well as audience and user demographic – meaning AR is naturally becoming more widespread.

Spark AR is an efficient and versatile way to create AR experiences and it widens the playing field – allowing anyone with an interest in AR to access essential functions.

CHAPTER 4: PROJECTS ON AR

4.1 Surprise Text

This project has been made in Spark AR Studio using Face Tracker Application.

Here in this project a text appears on the blinking of eyes and disappears while doing the same.

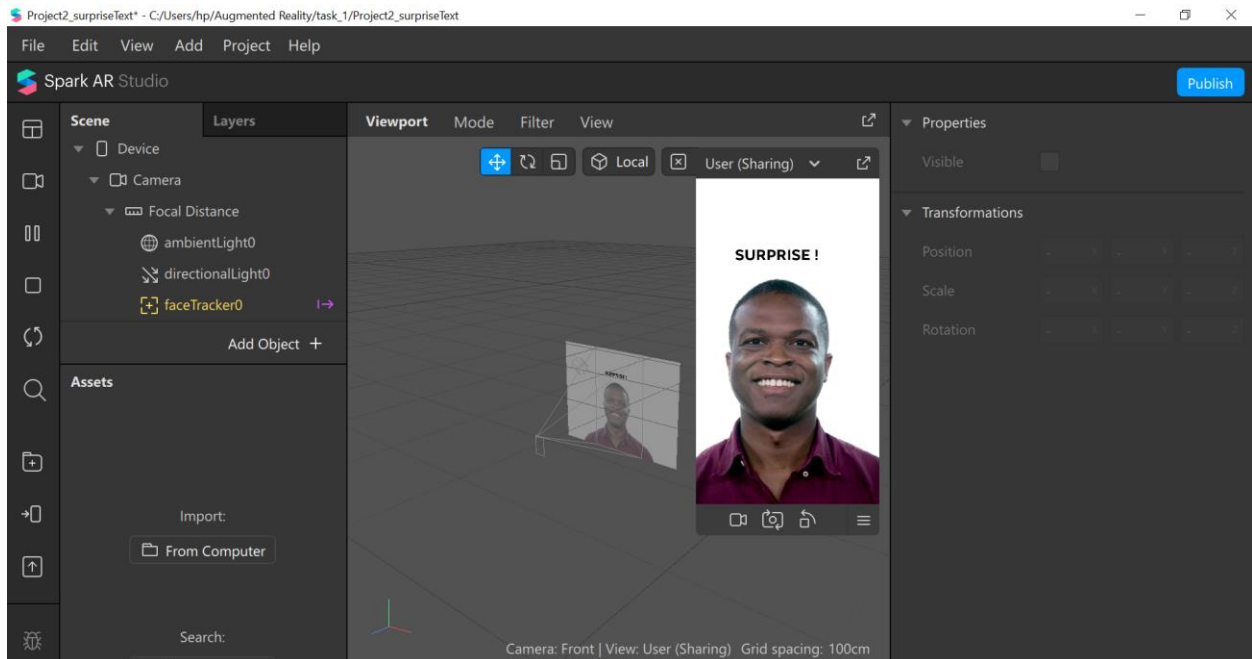


Figure 1: Surprise Text on Spark AR Studio

4.2 Changing Eye Colour

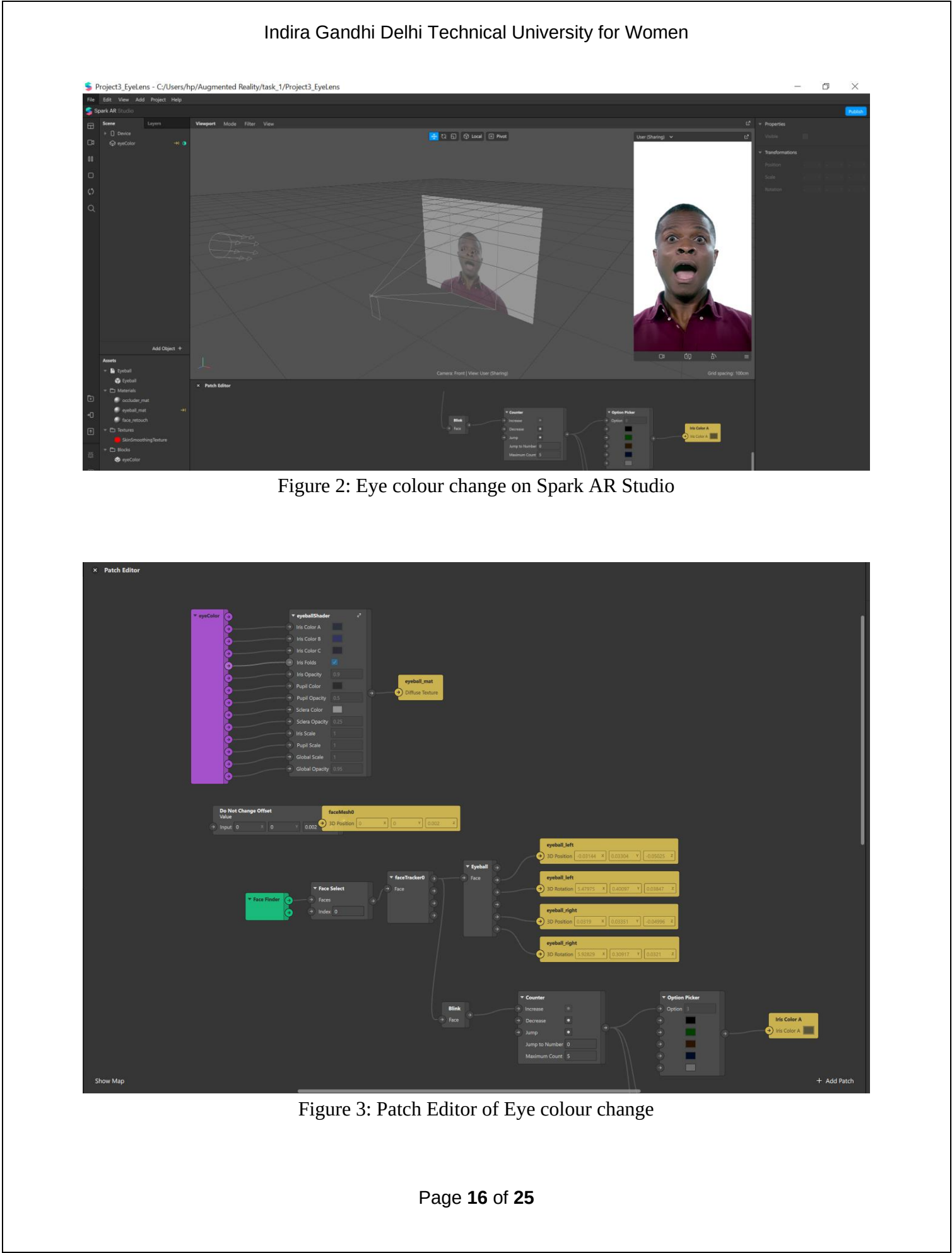
In this project, colour of iris changes while blinking of eyes. Various objects of Spark AR have been used while making this, like Face Tracker, Plane Tracker, Face Mesh, Ambient Light, counter etc.

Patch Editor is used to make logic of this project.

The screenshot shows the Spark AR Studio interface. The top menu bar includes File, Edit, View, Add, Project, and Help. The main viewport displays a 3D scene with a face mesh and a camera view. The Patch Editor at the bottom shows a sequence of patches for eye color change, including 'eyeColor', 'eyeColorShader', 'eyeColor_mat', 'faceMesh0', 'Face', 'eyeColor_left', 'eyeColor_right', 'Counter', and 'Option Picker'.

Figure 2: Eye colour change on Spark AR Studio

The screenshot shows the Spark AR Studio interface. The top menu bar includes File, Edit, View, Add, Project, and Help. The main workspace is divided into several panels: a left sidebar with 'Scene' and 'Assets' panels, a central 'Viewport' showing a 3D scene with a face mesh and a camera view, and a right sidebar with 'Properties' and 'Transformations' panels. The 'Patch Editor' is visible at the bottom, showing a sequence of patches for eye color change. The patches include 'eyeColor', 'eyeColorShader', 'eyeColor_mat', 'faceMesh0', 'Face Tracker0', 'Face', 'eyeColor_left', 'eyeColor_right', 'eyeColor_counter', and 'eyeColor_picker'.



The screenshot shows the Spark AR Studio interface. The top menu bar includes File, Edit, View, Add, Project, and Help. The main viewport displays a 3D scene with a face mesh and a camera view. The Patch Editor at the bottom shows a sequence of patches for eye color change, including 'eyeColor', 'eyeColorShader', 'eyeColor_mat', 'faceMesh0', 'Face', 'eyeColor_left', 'eyeColor_right', 'Counter', and 'Option Picker'.

Figure 2: Eye colour change on Spark AR Studio

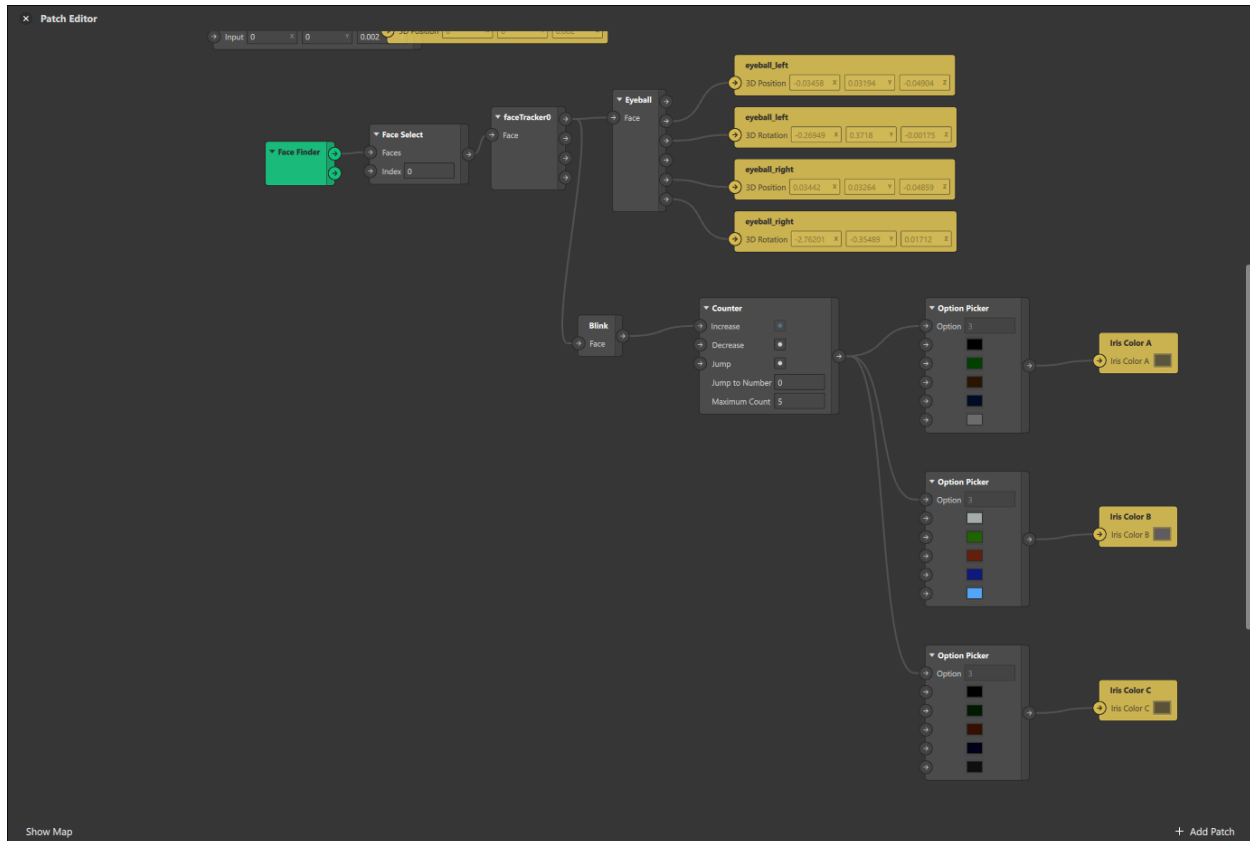


Figure 4: Patch Editor of Eye colour change

4.3 This or That Game

This project works in a similar way of This or That Game. In this, name of a pair of things appears and it changes the pair of things while we tilt our head left or right and it ends after a pair of names.

Logic of this game has been build using Patch Editor and objects of Spark AR Studio and learned the following things:

- How to use head rotation
- How to use logical patches
- How to use loop animation patch
- How to use utility patch
- Use of animation sequences

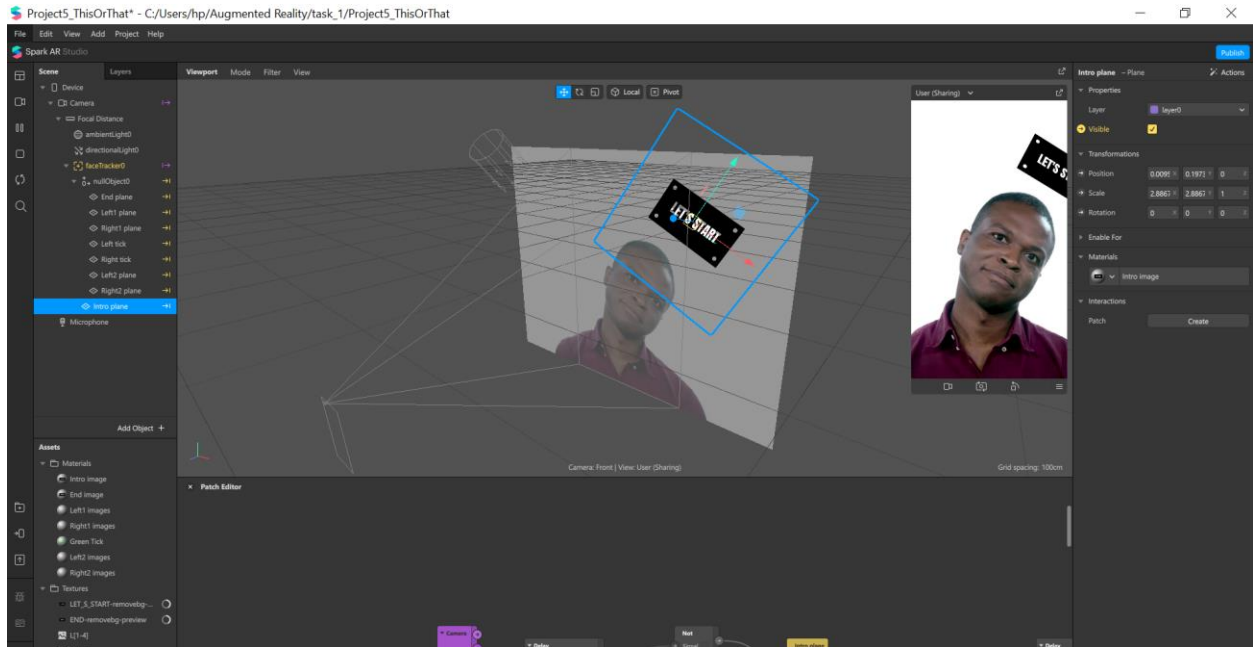


Figure 5: This or That game on Spark AR Studio

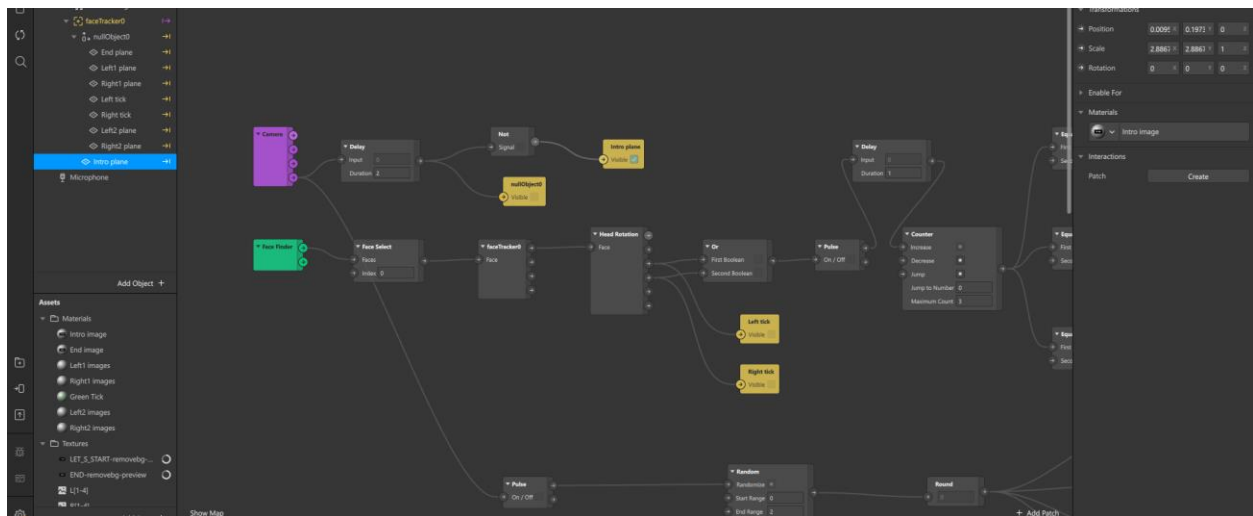


Figure 6: Patch Editor of This or That game

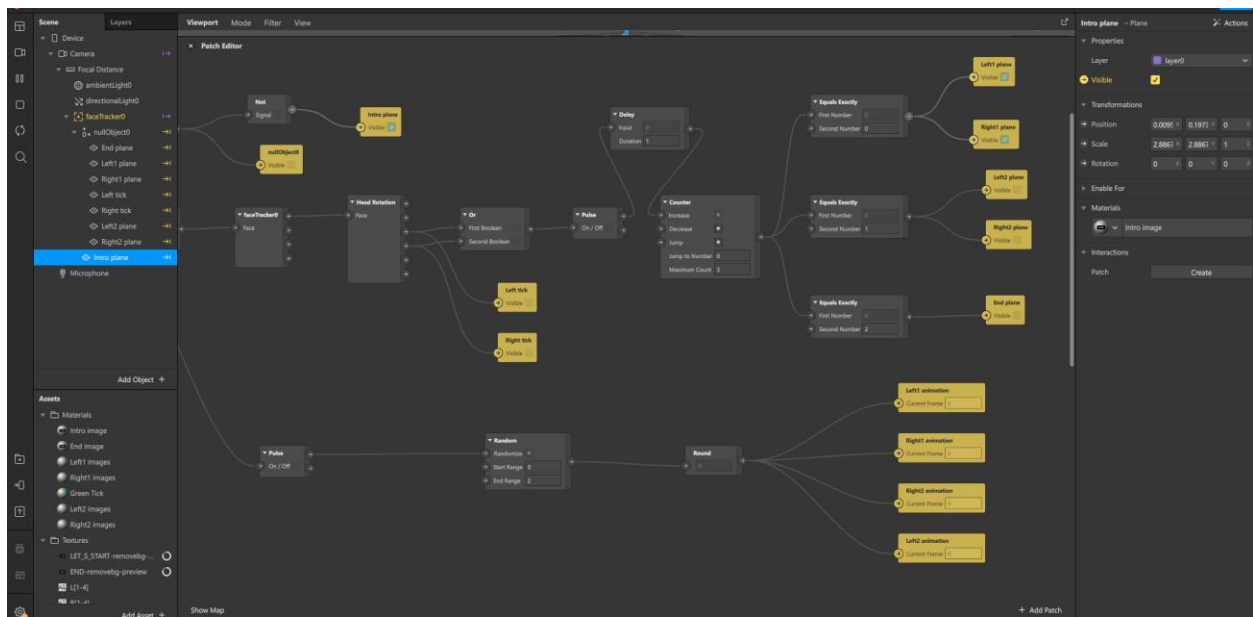


Figure 7: Patch Editor of This or That game

4.4 Guess the Emoji Game

This project works in a similar way of Guess the emoji game. Its logic has been built in a similar manner of above two projects using Patch Editor and objects of Spark AR studio. Loop animation Patch, Frame Transition Patch, Round Patch, Random Patch, Logical Patch has also been used.

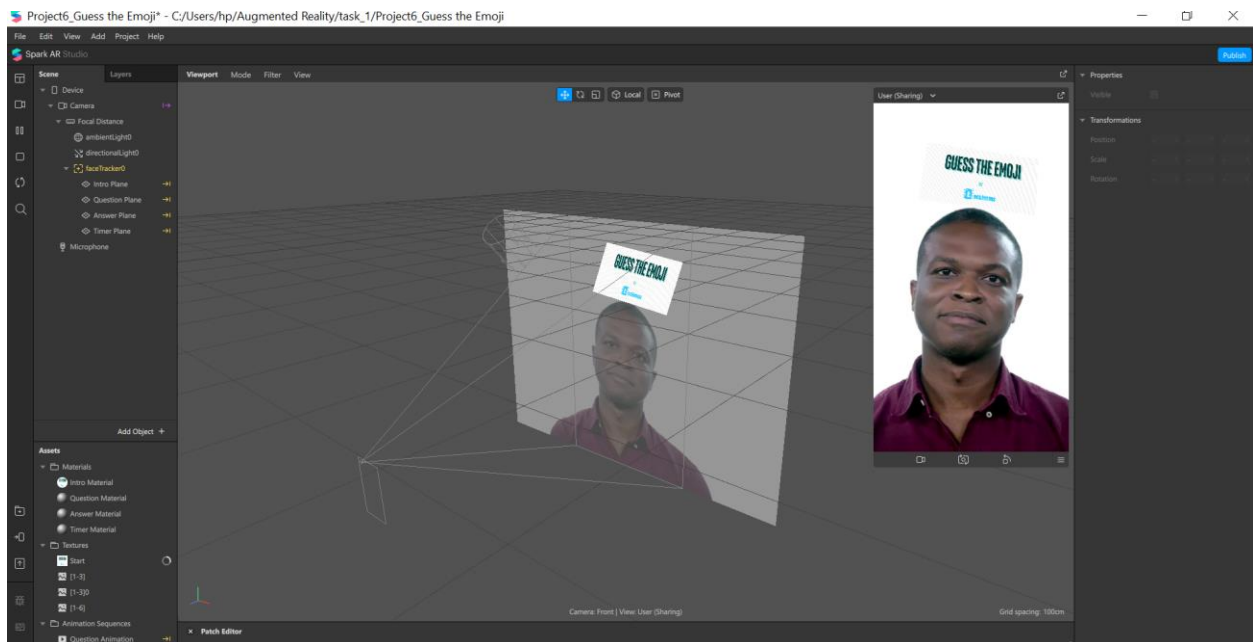
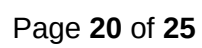


Figure 8: Guess the Emoji game on Spark AR Studio



CHAPTER 5: CONCLUSION

AR is a technology that lets people superimpose digital content over a real-world environment. AR got a lot of attention in 2016 when the game Pokémon Go made it possible to interact with Pokémon superimposed on the world via a smartphone screen.

AR is not a fully immersive experience like VR. While virtual reality requires users to don a special headset and pulls them into a completely digital world, AR lets them continue interacting with the physical world around them.

AR can be used in various fields apart from its limited scope to digital world, like education, entertainment, gaming and selling.

Today, most of these augmented reality experiences are made possible by smartphones. However, the development of more advanced AR devices (like Apple's AR glasses or Microsoft's HoloLens) could open the door to even more applications.

CHAPTER 6: FUTURE SCOPE

For all those who use mobile devices, AR is the future. It allows users to see a completely different reality.

The opportunities in AR are enormous, it's just that people need to sense these opportunities and make use of it in a manner that is beneficial to them.

Augmented reality has impacted every sphere of life. The major sectors are education, health and medical, gaming, marketing and advertisement industry.

Marketing and advertisement industry will explode with augmented reality devices. Face recognition, Advanced GPS systems and what not.

Think of a world where all the digital information is infused with the world around us, such as while taking a stroll in the park, messages from our friends pop-up in the corner of our field of vision or may be if we lose our way back home, a voice instruction can be issued to navigate us and have direction arrows appearing as if they are markings on the pavement. Oh, and if we come across a person whom we recognise but can't figure out where from? No problem, facial recognition software comes to the rescue which will pull up their twitter feed and give the required details.

Education industry will have tremendous impact with augmented reality in place. Augmented reality creates an engaging environment for students and make harder concepts easy to understand and learn. Moreover, it makes learning interesting and interactive.

Augmented reality plays a crucial rule in the future of **Health** and **medical industry**. Surgeries would be performed flawlessly reducing the chances of any default. Also, less time would be taken to perform these surgeries.

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